

Generation Interconnection System Impact Study Report

For Queue Project Q21-1920

Lighthouse Hill 345 kV

May 18, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1920. The Interconnection Customer proposes a 130.2 MW solar photovoltaic facility interconnecting at the Lighthouse Hill 345 kV substation in the Pennsylvania area. The request was received on October 31, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 555 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$266,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: NORTH - SHEPAUG 138 kV

Contingency: RITTER PARK - SUGAR HILL 345 kV line

Loading: 100.3% (Rating: 372 MVA)

DFAX: 5.2%

Pre-project loading: 98.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NORTH - SHEPAUG 138 kV	Reconductor line	\$1,731,000	15% (\$266,000)
	Total allocated to project		\$266,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.